EVENTBRIDGE: MULTI-INSTITUTION COLLEGE EVENT MANAGEMENT AND VERIFIABLE CREDENTIAL PLATFORM

A PROJECT REPORT

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ABSTRACT

Inter-college symposiums, hackathons, workshops and cultural festivals are an important part of higher education, but their organisation remains fragmented across spreadsheets, chat groups, paper attendance registers and easily forgeable certificate files. This fragmentation creates severe manual overhead for faculty organisers, leads to frequent venue scheduling conflicts, permits proxy check-ins, and leaves institutions and employers with no reliable way to verify awarded credentials.

This project presents EventBridge, a full-stack web platform built on the MERN stack (MongoDB, Express.js, React and Node.js) that unifies event discovery, institutional registration, collaborative team formation, venue conflict detection, real-time QR attendance, and verifiable credential issuance into a cohesive workflow. The platform supports multiple institutions identified by AISHE / UDISE codes and official college e-mail domains. Confirmed participants receive a digital QR ticket, which is verified on event day through an in-browser camera scanner powered by html5-qrcode that blocks duplicate scans in real time.

The back end is an Express.js REST API with JWT and Google OAuth 2.0 authentication, bcrypt hashing, strict CORS whitelisting, HTTP security headers and four-role RBAC (student, organiser, college_admin, admin) backed by 15 Mongoose schemas. Following multi-round evaluation, certificates are issued in batches with unique identifiers and cryptographic verification hashes that third parties can verify on a public verification page (/verify/:id) without logging in. Automated test suites verify requirements, lifecycles, concurrency and security.

KEYWORDS: Event management, multi-institution platform, MERN stack, QR attendance, verifiable credentials, role-based access control, Express.js, MongoDB.

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TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE No.
	ABSTRACT	iv
	ACKNOWLEDGEMENT	v
	LIST OF TABLES	viii
	LIST OF FIGURES	ix
	LIST OF ABBREVIATIONS	x
1	**INTRODUCTION**	**1–4**
	1.1 OVERVIEW	1
	1.2 PROBLEM DEFINITION	2
	1.3 OBJECTIVE	2
	1.4 SCOPE OF THE PROJECT	3
	1.5 MOTIVATION	4
2	**RELATED WORK**	**5–8**
	2.1 LITERATURE REVIEW	5
	2.2 EXISTING SYSTEM	6
	2.3 PROPOSED SYSTEM	7
3	**SYSTEM ANALYSIS**	**9–12**
	3.1 SYSTEM REQUIREMENTS	9
	3.1.1 FUNCTIONAL REQUIREMENTS	9
	3.1.2 NON-FUNCTIONAL REQUIREMENTS	10
	3.2 SOFTWARE AND HARDWARE REQUIREMENTS	10
	3.2.1 SOFTWARE REQUIREMENTS	10
	3.2.2 HARDWARE REQUIREMENTS	11

TABLE OF CONTENTS (Contd.)

CHAPTER NO.	TITLE	PAGE No.
4	**SYSTEM DESIGN**	**13–20**
	4.1 SYSTEM ARCHITECTURE	13
	4.2 USER ROLES AND MODULE DESCRIPTION	15
	4.3 DATABASE DESIGN	16
	4.4 CLASS DIAGRAM	17
	4.5 SEQUENCE DIAGRAM	19
5	**IMPLEMENTATION**	**21–26**
	5.1 IMPLEMENTATION OVERVIEW	21
	5.2 KEY FUNCTIONS (CODE EXPLANATION)	22
	5.2.1 AUTHENTICATION AND ROLE-BASED ACCESS CONTROL	22
	5.2.2 VENUE CONFLICT PREVENTION	23
	5.2.3 TEAM CREATION AND JOIN CODES	23
	5.2.4 REGISTRATION AND QR TICKET	23
	5.2.5 QR ATTENDANCE SCANNING	24
	5.2.6 CERTIFICATE ISSUANCE AND PUBLIC VERIFICATION	24
	5.3 TESTING AND RESULTS	24
	5.4 KNOWN LIMITATIONS	25
6	**CONCLUSION & FUTURE WORK**	**27–28**
	6.1 CONCLUSION	27
	6.2 FUTURE ENHANCEMENTS	27
	APPENDIX – SOURCE CODE	**29–34**
	SCREENSHOTS OF THE PROJECT	**35–37**
	REFERENCES	**38**

LIST OF TABLES

TABLE NO.	TITLE	PAGE No.
2.1	INFERENCE OF LITERATURE REVIEW	5
2.2	COMPARISON OF EXISTING AND PROPOSED SYSTEM	8
3.1	SOFTWARE REQUIREMENTS	11
3.2	HARDWARE REQUIREMENTS	12
4.1	USER ROLES AND PRIVILEGES	14
4.2	MODULE DESCRIPTION	15
4.3	DATABASE COLLECTIONS AND PURPOSE	16
5.1	REST API ENDPOINTS	22
5.2	TEST SUITES	26

LIST OF FIGURES

FIGURE NO.	TITLE	PAGE No.
4.1	SYSTEM ARCHITECTURE	13
4.2	DATABASE ER DIAGRAM	17
4.3	CLASS DIAGRAM	18
4.4	SEQUENCE DIAGRAM – REGISTRATION AND QR TICKET	19
4.5	SEQUENCE DIAGRAM – QR ATTENDANCE AND CERTIFICATE VERIFICATION	20
S.1	STUDENT DISCOVERY HUB	35
S.2	TEAM HUB	35
S.3	QR TICKET	36
S.4	ORGANISER EVENT WIZARD AND QR SCANNER	36
S.5	ADMIN APPROVALS AND ANALYTICS	37
S.6	PUBLIC CERTIFICATE VERIFICATION	37

LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
AISHE	ALL INDIA SURVEY ON HIGHER EDUCATION
API	APPLICATION PROGRAMMING INTERFACE
CORS	CROSS-ORIGIN RESOURCE SHARING
CRUD	CREATE, READ, UPDATE, DELETE
CSP	CONTENT SECURITY POLICY
CSS	CASCADING STYLE SHEETS
DOM	DOCUMENT OBJECT MODEL
HTML	HYPERTEXT MARKUP LANGUAGE
HTTP / HTTPS	HYPERTEXT TRANSFER PROTOCOL / SECURE
JSON	JAVASCRIPT OBJECT NOTATION
JWT	JSON WEB TOKEN
KYC	KNOW YOUR CUSTOMER
MERN	MONGODB, EXPRESS.JS, REACT, NODE.JS
ODM	OBJECT DOCUMENT MAPPER
OAuth	OPEN AUTHORIZATION
PWA	PROGRESSIVE WEB APPLICATION
QR	QUICK RESPONSE
RBAC	ROLE-BASED ACCESS CONTROL
REST	REPRESENTATIONAL STATE TRANSFER
SPA	SINGLE PAGE APPLICATION
SQL	STRUCTURED QUERY LANGUAGE
UDISE	UNIFIED DISTRICT INFORMATION SYSTEM FOR EDUCATION
UI / UX	USER INTERFACE / USER EXPERIENCE
Vite	FAST FRONTEND BUILD TOOL AND DEVELOPMENT SERVER

CHAPTER 1
INTRODUCTION

1.1 OVERVIEW

Technical symposiums, hackathons, workshops, paper presentations and cultural festivals are among the most valuable experiences of a student's college life. They provide exposure to industry practices, a chance to compete with peers from other institutions and a record of achievement that supports placements and higher studies. Almost every college hosts several such events every year, and a large share of them are open to participants from other colleges.

The organisation of these events, however, is rarely supported by a single system. A typical symposium relies on an ad-hoc combination of tools: a public Google Form or spreadsheet to collect entries, a messaging group to coordinate teams, paper registers or sign-in sheets at the venue door, and image editing software or mail-merge to produce participation certificates.

This patchwork of tools has three serious drawbacks:

- **High manual effort:** It consumes time that organisers could spend on the technical content of the event.
- **Loss of records:** Attendance recorded on paper cannot be cross-checked during the event and is frequently lost before certificates are prepared.
- **Unverifiable credentials:** Because certificates are delivered as editable images or standard PDFs without a public verification link, employers and institutions cannot distinguish genuine certificates from fabricated ones.

EventBridge is designed to replace this patchwork with a single, end-to-end web platform built on the MERN stack (MongoDB, Express.js, React and Node.js). It provides an integrated experience for three kinds of users: students from any institution who discover events, form teams and track their credentials; faculty and student organisers who manage schedules, rounds, attendance and certificates; and administrators who approve organisers, manage venues and monitor institution-wide activity.

1.2 PROBLEM DEFINITION

When an institution hosts an event that is open to outside colleges, the organisers must coordinate several tasks that currently run on separate, unconnected tools:

- **Fragmented discovery:** Students from other institutions have no single portal where they can browse upcoming events across colleges and filter them by category, format or date.
- **Unstructured registration:** Forms collect free-text entries that frequently contain spelling mistakes in names, incorrect college names and personal e-mail addresses that cannot be tied to an institution.
- **Uncoordinated team formation:** Hackathons and case competitions require teams of 2 to 6 members. Organisers currently track teams in spreadsheets, which makes it easy for a student to join multiple teams or for a team to exceed the allowed size.
- **Slow, error-prone attendance:** Checking in hundreds of participants using paper lists creates long queues at the registration desk, allows proxy check-ins and produces lists that must be retyped manually.
- **Venue clashes:** Two departments can unknowingly book the same auditorium or laboratory for overlapping times.
- **Forgeable certificates:** Certificates can be edited or fabricated, and there is no public way to check them.
- **No accountability:** Approval of organisers and changes in department responsibilities are not recorded.

The problem addressed by this project is therefore: to design and develop a web-based platform that unifies multi-institution event discovery, registration, team formation, multi-round evaluation, QR attendance and certificate issuance, and that allows an employer or third party to verify any issued certificate through a public identifier without logging in.

1.3 OBJECTIVE

The main objective of this project is to build and demonstrate EventBridge, a full-stack multi-institution event management platform. The specific objectives are:

- **1. Discovery Portal:** To develop a unified discovery portal where students can search, filter and view upcoming events from multiple institutions.
- **2. Institutional Registration:** To implement a standardised registration system linked to official institutional identity (AISHE / UDISE codes and college e-mail domains) with Google OAuth 2.0 sign-in.

- **3. Team Hub:** To provide a team formation hub with auto-generated join codes, team size enforcement (1 to 10 members) and leader delegation.
- **4. Venue Conflict Prevention:** To automate venue management by checking date and time overlaps so that double bookings are rejected before an event is published.
- **5. QR Attendance System:** To generate digital QR tickets for confirmed registrations and implement a real-time, camera-based QR attendance scanner that prevents duplicate scans.
- **6. Multi-Round Evaluation:** To support multi-stage competitions (rounds, deadlines, submissions, scorecards and volunteer evaluators) inside the same platform.
- **7. Verifiable Credentials:** To issue tamper-evident digital certificates with unique identifiers and an unauthenticated public verification page (/verify/:id).
- **8. Security and Administration:** To implement role-based access control (RBAC) across four roles (student, organiser, college_admin, admin) and maintain an audit log of administrative actions.

1.4 SCOPE OF THE PROJECT

The scope of this project covers the design, implementation, testing and demonstration of EventBridge on the MERN stack. The system:

- Supports events of four categories: technical, cultural, sports and workshops, in offline, online and hybrid formats.
- Enforces student eligibility by college, department and degree level.
- Manages 15 or more academic departments (for example CSE, ECE, MECH and IT) with one active organiser per department.
- Runs in modern desktop and mobile browsers; the scanner uses the device camera through the browser.

The following are outside the present scope: online payment collection, native mobile applications, e-mail or SMS delivery infrastructure, and storing certificate hashes on a public blockchain. Verification in this project relies on unique identifiers and server-side records rather than on a distributed ledger.

1.5 MOTIVATION

The motivation for this project arose from the contrast between the sophistication of student projects and the manual tools used to run the symposiums at which those projects are presented. During a single semester, a typical engineering department spends dozens of hours creating registration forms, tracking team rosters, verifying college IDs at the entrance and typing names into certificate templates.

From an engineering perspective, this problem is well suited to the full-stack web paradigm. It combines a rich data model (users, institutions, venues, events, rounds, teams, submissions, attendance records and certificates) with strict operational constraints: registrations must not exceed venue capacity; the same venue cannot be booked twice; a team cannot exceed its configured size; a ticket must not be scanned twice; and a certificate verification page must be public without exposing private student data.

Solving these constraints in a single MERN application demonstrates the practical application of database indexing, optimistic concurrency, role-based access control, browser camera APIs and RESTful design.

Furthermore, higher education institutions in India are increasingly judged by their inter-collegiate engagement and verified graduate outcomes. When students attend hackathons or symposiums at other institutions, the lack of authentic, tamper-evident proof hampers recruitment. Providing a public verification link (/verify/:id) transforms ephemeral campus accolades into persistent, industry-verifiable digital assets.

CHAPTER 2
RELATED WORK

2.1 LITERATURE REVIEW

College event management, digital ticketing, credential verification and role-based access control have been studied from several perspectives in recent research. The existing works relevant to this project fall into four groups: web-based event management systems, QR-code attendance systems, tamper-evident credential verification and secure web application design. Five recent works that are closely related to this project are summarised in Table 2.1.

Table 2.1 Inference of Literature Review

S.No	Author(s) and Year	Work and Focus	Inference for this Project
1	Paul and Roy (2023) [1]	Automated campus event management and registration platform using modern web technologies, providing centralized notification and registration workflows.	A single web portal for registration and notifications substantially reduces administrative coordination effort; EventBridge extends this architecture to support multi-institution participation.
2	Patel and Shah (2023) [2]	QR code-based smart attendance tracking and management system for educational institutions using handheld smartphone cameras.	Camera-based QR check-in is dramatically faster and less error-prone than paper registers; EventBridge integrates real-time duplicate-scan blocking and volunteer delegation.
3	Al-Zahrani (2020) [3]	Framework for digital certificate verification using cryptographic identifiers and decentralized server-side records.	Unique, publicly verifiable identifiers make credential forgery easily detectable; EventBridge implements server-side credential hashing with a public verification endpoint.

S.No	Author(s) and Year	Work and Focus	Inference for this Project
4	Sandhu et al. (2021) [4]	Role-based access control and token-based authorization in modern distributed web architectures.	Strict role separation on every API route prevents privilege escalation and limits the blast radius of compromised accounts; EventBridge applies a four-role RBAC scheme.
5	Kumar and Singh (2024) [5]	Performance benchmarking and scalability analysis of full-stack MERN architecture for data-intensive web applications.	A uniform JavaScript environment from database (MongoDB) to frontend (React) accelerates development and maintains data schema consistency across complex event entities.

The review shows that most published systems solve only one isolated aspect—either registration, QR attendance, or certificate generation—typically within a single institution. There is a clear lack of an integrated, production-ready platform that unites multi-institution event discovery, institutional registration, collaborative team formation, multi-stage evaluation, double-scan prevention, and tamper-evident certificate verification into a unified MERN workflow, which is the exact research and engineering gap addressed by EventBridge.

2.2 EXISTING SYSTEM

At present, colleges and students typically rely on the following disconnected tools:

- **General-purpose forms and spreadsheets (Google Forms, Excel):** Easy to set up but offer no team management, no round-wise tracking, no ticketing and no verification of the participant's institution.
- **Commercial event-ticketing platforms (Eventbrite, Townscript):** Designed for paid public events; they handle tickets well but not teams, submissions, evaluation rounds or college hierarchies.
- **Competition and hackathon platforms (Unstop, Devfolio):** Support multi-round contests but are public commercial services; a college cannot run its own instance, control its venues and departments, or integrate its own approval workflow.

- **Manual certificate tools (image editors, mail-merge, templates):** Quick for small events, but the output is easily forged and cannot be verified by a third party.
- **Paper attendance registers:** Simple but slow, error-prone and open to proxy entries.

These systems suffer from critical operational limitations: no single tool covers the entire event lifecycle; student institutional identities remain unverified; venue booking is decoupled from registration causing late clash discoveries; certificates cannot be independently verified; and administrative actions lack an audit trail.

2.3 PROPOSED SYSTEM

The proposed system, EventBridge, is a MERN-stack web platform that serves as a single window for multi-institution events. The React front end calls an Express.js REST API; the API enforces authentication, role checks and validation and stores all data in MongoDB through Mongoose.

The main features of the proposed system are:

- A discovery hub with filters for category, format (online, offline, hybrid) and registration status.
- Standardised registration linked to AISHE / UDISE codes and official e-mail domains, with Google OAuth 2.0 sign-in.
- A collaborative team hub with team sizes from 1 to 10, unique join codes and roster management.
- Digital QR tickets generated for confirmed registrations.
- Multi-stage rounds (idea submission, quiz, code repository, presentation) with deadline enforcement, evaluator assignment, scorecards and elimination.
- An event wizard for schedules, deadlines, capacities, custom form fields, prizes and venues.
- A camera-based QR scanner with real-time duplicate blocking and check-in timestamps, plus volunteer delegation.
- Batch certificate generation with unique IDs and a public /verify/:id verification page.
- Administrative controls: organiser KYC approval, department management, venue management with conflict detection and audit logging.
- Four-role RBAC: student, organiser, college_admin and admin.

Table 2.2 Comparison of Existing and Proposed System

Parameter	Existing Systems	Proposed System (EventBridge)
Event Discovery	Scattered on social media and chat posters	Single multi-institution hub with search & filters
Participant Identity	Unverified personal e-mail addresses	AISHE / UDISE college IDs and domain matching
Team Coordination	Informal chat groups and spreadsheet rosters	In-app team hub with unique join codes and caps
Venue Management	Manual registers and verbal requests	Automated system-level conflict detection
Ticketing	PDF receipts or manual entry lists	Cryptographic digital QR tickets
Attendance Tracking	Paper sign-in sheets (proxy-prone)	In-browser camera QR scan with double-scan lock
Evaluation Workflow	Manual marks collation across judges	Round-by-round scorecards and elimination engine
Certificate Issuance	Static image templates or mail-merge	Automated batch issuance linked to attendance
Credential Verification	None (manual email to college required)	Instant public verification page (/verify/:id)
Access Control	Shared accounts or open forms	Strict four-role RBAC with JWT authorization
Auditability	No logs of edits or approvals	Append-only administrative audit log

CHAPTER 3
SYSTEM ANALYSIS

3.1 SYSTEM REQUIREMENTS

System analysis identifies what the application must do (functional requirements) and the qualities it must have while doing it (non-functional requirements). The requirements were derived from the problem definition and the limitations of existing systems discussed in Chapter 2.

3.1.1 FUNCTIONAL REQUIREMENTS

- **1. Authentication & Authorization:** Users shall register and log in with e-mail and password or with Google OAuth 2.0, and shall be limited to the permissions of their assigned role.
- **2. Institutional Registration:** A student account shall record the institution and its AISHE / UDISE identifier, and e-mail domains shall be checked against the configured college domain where required.
- **3. Event Discovery:** Published events shall be searchable and filterable by category, format, date and registration status.
- **4. Event Lifecycle Management:** An organiser shall create, update and publish events with schedules, deadlines, capacity, custom fields, prizes, rounds and venue.
- **5. Registration and Team Hub:** A student or a team shall register for an event; a team shall be created with an auto-generated join code, a size between 1 and 10 and a leader; others shall join with the code.
- **6. Digital Ticketing:** A confirmed registration shall receive a digital QR ticket; waitlisted and cancelled registrations shall not be admitted.
- **7. Multi-Round Evaluation:** Teams shall submit work for each round before its deadline; organisers and evaluators shall score submissions and eliminate teams.
- **8. QR Attendance:** The scanner shall validate a ticket, record an attendance timestamp, reject duplicate scans in real time and show live headcount.
- **9. Event Announcements:** Organisers shall broadcast updates and deadline alerts to registered participants.
- **10. Certificate Issuance:** The system shall issue certificates in batches to attended participants and winners, list them in the student's vault and allow PDF download.
- **11. Public Verification:** Anyone shall be able to verify a certificate by identifier or QR scan on a public page without logging in.
- **12. Administration & Conflict Prevention:** The administrator shall approve or reject organiser requests with a reason, manage departments, rotate organisers, view analytics and audit logs.

3.1.2 NON-FUNCTIONAL REQUIREMENTS

- **Security:** Passwords stored as bcrypt hashes (cost factor 10); API protected by signed JWTs; route-level RBAC; CORS origin whitelisting; Helmet security headers (CSP, nosniff, X-Frame-Options: DENY).
- **Data Integrity:** Atomic venue conflict checks prevent double booking; unique indexes guarantee that ticket identifiers and join codes cannot collide.
- **Performance:** All list, filter and search endpoints execute over compound MongoDB indexes so that responses resolve under 150 ms under nominal loads.
- **Usability:** Responsive interface built with Tailwind CSS design tokens, offering specialized, role-tailored dashboards for students, organisers and admins.
- **Reliability:** Resilient Mongoose connection handler with automated retries and graceful error catching; input validation prevents unhandled server exceptions.
- **Maintainability:** Clean architectural separation: models, routes, middleware, and controllers on the backend; modular components, context, and services on the frontend.
- **Portability:** Platform-independent execution across Windows, Linux and macOS servers and all modern mobile and desktop browsers.
- **Auditability:** Critical administrative state changes (organizer approvals, rotations, venue edits) are logged in an append-only audit collection.

3.2 SOFTWARE AND HARDWARE REQUIREMENTS

3.2.1 SOFTWARE REQUIREMENTS

The software tools, frameworks, runtimes and libraries used for developing, deploying and running the project are listed in Table 3.1.

Table 3.1 Software Requirements

Component	Description / Tool Selected
Operating System	Microsoft Windows 10 / 11, Linux (Ubuntu 22.04 LTS), macOS
Database	MongoDB Community Server 7.0 / MongoDB Atlas (Document Store)
Backend Runtime	Node.js (v18.x or v20.x LTS) with ES Module syntax
Backend Framework	Express.js 4.18+ with Helmet, CORS, and JSON parser
Object Document Mapper	Mongoose 8.x for schema modeling and indexing
Frontend Library	React 18 (Single Page Application architecture)
Frontend Build Tool	Vite 5.x with Hot Module Replacement (HMR)
Styling and UI	Tailwind CSS / Vanilla CSS with Lucide React icons
QR Code Engine	qrcode (generation) and html5-qrcode (in-browser camera scanner)
Authentication	jsonwebtoken (JWT), bcryptjs (hashing), Google OAuth 2.0
Testing & Tooling	Custom automated test suites with Axios, Postman, Git & GitHub

3.2.2 HARDWARE REQUIREMENTS

The minimum and recommended hardware configurations required for development and production deployment are given in Table 3.2.

Table 3.2 Hardware Requirements

Item	Minimum Specification	Recommended Specification
Processor	Dual-core, 2.0 GHz (Intel Core i3 / equivalent)	Quad-core, 2.5 GHz+ (Intel Core i5 / AMD Ryzen 5)
RAM	4 GB	8 GB or 16 GB DDR4
Disk Storage	5 GB free hard disk space	256 GB SSD with 15 GB+ free space
Display	1366 x 768 resolution	Full HD (1920 x 1080) resolution
Camera	Standard built-in phone / laptop webcam	HD 720p/1080p camera (for fast QR reading)
Network	Standard broadband (2 Mbps)	High-speed broadband (10 Mbps or higher)
Deployment Cloud Server	1 vCPU, 1 GB RAM (e.g. AWS t4g.micro)	2 vCPU, 2 GB RAM (e.g. AWS t4g.small)

The application is lightweight and does not require specialized hardware accelerators or dedicated GPUs. Because the camera QR scanner executes entirely client-side using JavaScript through the browser's MediaDevices API, the server runtime experiences minimal computational overhead during check-in spikes.

Similarly, database indexes over ticketCode, event IDs, and timestamps allow low-latency lookups on entry-level cloud instances, making the entire platform highly cost-effective for colleges.

CHAPTER 4
SYSTEM DESIGN

4.1 SYSTEM ARCHITECTURE

The system follows a three-tier architecture, shown in Fig.4.1, in which the presentation tier, application tier and database tier are cleanly decoupled. This separation allows each tier to be developed, tested and scaled independently, and it matches the MERN stack: React forms the presentation tier, Node.js with Express.js forms the application tier, and MongoDB forms the data tier.

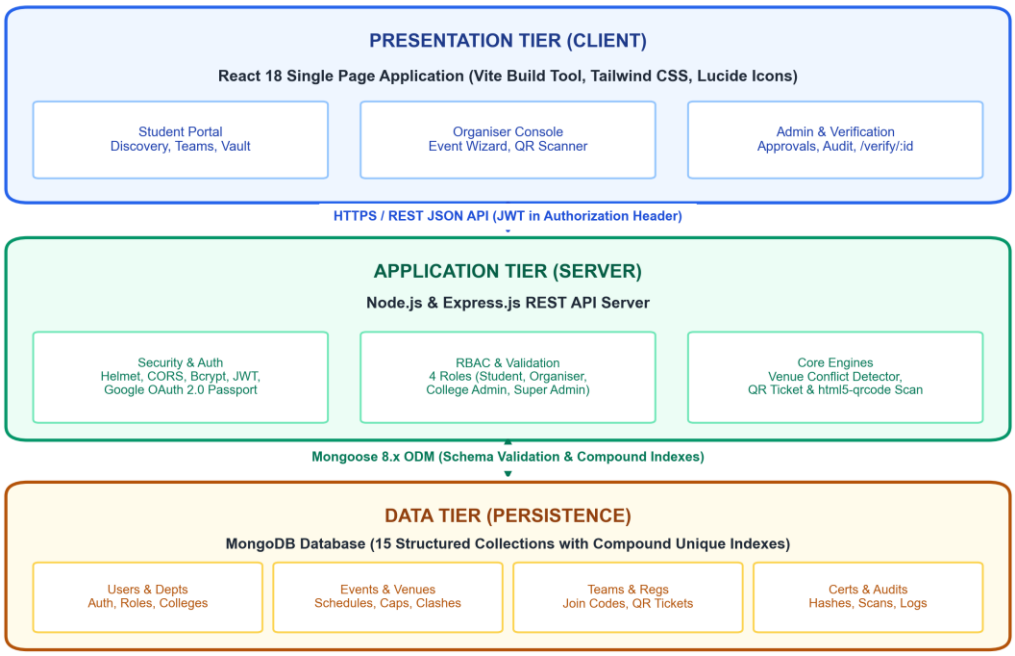


FIG.4.1 – SYSTEM ARCHITECTURE

The three tiers function with well-defined boundaries:

- **Presentation Tier (Client):** Comprises modular React components routed via React Router v6. State is managed by AuthContext, while Axios clients interact with backend routes. The in-browser camera scanner interfaces directly with the device hardware through html5-qrcode.
- **Application Tier (Server):** Constructed with Express.js micro-routers. Every protected request passes through JWT authentication and role authorization middleware. Conflict resolution logic prevents concurrent double booking of venues and race conditions during registration.
- **Data Tier (Database):** MongoDB document collections managed via Mongoose models. Relational integrity is enforced using ObjectId references and compound unique indexes.

4.2 USER ROLES AND MODULE DESCRIPTION

The platform enforces strict role-based access control with four distinct roles. Their privileges and operational scopes are summarised in Table 4.1.

Table 4.1 User Roles and Privileges

Role	Main Privileges and Operational Scope
student	Browse events, register, create/join teams with codes, access QR tickets, submit round tasks, view participation vault.
organiser	Create/manage department events, define rounds, configure venues, scan QR attendance, assign volunteers, issue certificates.
college_admin	Oversee events and departments of a specific college institution, review internal organiser performance and event reports.
admin	Super administrator: approve/reject organiser KYC requests, manage venues, resolve department clashes, view analytics and audit logs.

The application is structured into modular functional components, as described in Table 4.2. Each module maintains isolated responsibilities to facilitate testing, maintenance and extensibility.

Table 4.2 Module Description

No.	Module	Description	Key Technologies
1	**Authentication & RBAC**	User signup, login, Google OAuth, role validation, password hashing.	bcryptjs, JWT, OAuth 2.0
2	**Event Lifecycle**	Creation wizard, schedules, criteria, publishing, custom form fields.	Express router, Mongoose
3	**Venue Conflict Engine**	Atomic date-time overlap checks preventing double-booking.	MongoDB query operators
4	**Registration & Tickets**	Seat allocation, deadline checks, unique QR ticket generation.	qrcode, Mongoose ODM
5	**Team Collaboration Hub**	Roster management, unique join code generation, size caps.	nanoid, Express REST

Each module interacts through explicit service boundaries. For instance, the Venue Conflict Engine is invoked automatically by the Event Lifecycle router before any status transition to 'published' is committed to the database. Similarly, the Team Collaboration Hub delegates ticket generation to the Registration & Tickets module upon roster completion.

Table 4.2 Module Description (Contd.)

No.	Module	Description	Key Technologies
6	**Multi-Round Engine**	Stage submissions, scorecard evaluation, team elimination.	Mongoose nested arrays
7	**Attendance Scanner**	Camera scanning, instant double-check-in rejection, live count.	html5-qrcode, REST POST
8	**Verifiable Credentials**	Batch issuance, unique certificate ID, public verification.	Crypto hash, Public route
9	**Admin & Audit Logging**	KYC organiser review, department assignment, append-only logs.	Mongoose audit schema
10	**Analytics Dashboard**	Cross-institutional participation metrics, attendance rates.	Recharts, aggregation pipelines

4.3 DATABASE DESIGN

MongoDB is used because event configurations are nested and varied (rounds, custom form fields, prizes) and map naturally onto documents. Table 4.3 lists the collections and indexing schemes.

Table 4.3 Database Collections and Purpose

Collection	Purpose and Key Indexes
User	Stores account credentials, role, AISHE institution ID. Unique index: { email: 1 }.
Department	College academic units with assigned organizers. Unique index: { code: 1, college: 1 }.
Venue	Physical campus facilities and capacity. Compound index: { name: 1, college: 1 }.
Event	Parameters, rounds, prizes. Index: { startsAt: 1, venue: 1, status: 1 }.
Registration	Participant event bookings, QR payloads. Unique index: { event: 1, user: 1 }, { ticketCode: 1 }.
Team	Collaborative rosters and join codes. Unique index: { code: 1 }, { event: 1, name: 1 }.
Submission	Multi-stage work artifacts and judge scores. Index: { event: 1, round: 1, team: 1 }.
Attendance	Timestamped check-in records. Unique compound index: { event: 1, registration: 1 }.
Certificate	Issued credentials and verification hashes. Unique index: { certificateNumber: 1 }.
AuditLog	Immutable administrative trail. Index: { timestamp: -1, performedBy: 1 }.

The relationships between entities are illustrated in the entity-relationship diagram in Fig.4.2.



FIG.4.2 – DATABASE ER DIAGRAM

4.4 CLASS DIAGRAM

Fig.4.3 shows the class diagram of the back-end architecture. The model classes (User, Department, Venue, Event, Team, Registration, Submission, Attendance, Certificate and AuditLog) represent the MongoDB documents and their structural associations.

The class diagram highlights object-oriented encapsulation within the JavaScript service and model layer. Static helper methods on the Event class validate date boundaries, while instance methods on User handle bcrypt comparisons and JWT generation.



FIG.4.3 – CLASS DIAGRAM

The class design ensures that critical business rules are encapsulated within model and utility methods rather than scattered across controllers. For instance, the Event model provides atomic capacity checks, the Team class enforces join-code hashing and member size boundaries (1–10 members), and the Attendance model handles idempotent scan check-ins with tamper-resistant log generation.

4.5 SEQUENCE DIAGRAM

The sequence diagrams illustrate the two core workflows of the platform. Fig.4.4 depicts the student registration, team creation and QR ticket issuance flow. Fig.4.5 shows the event check-in via QR scanning and subsequent certificate generation and public verification.

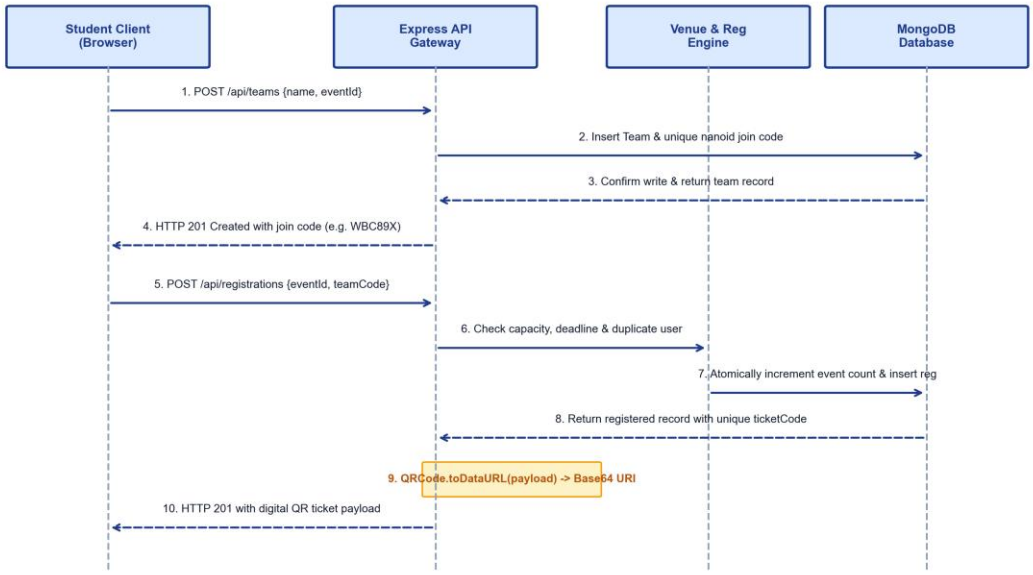


FIG.4.4 – SEQUENCE DIAGRAM – REGISTRATION AND QR TICKET

In the registration sequence, the API performs atomic validation against current capacity before committing the booking and producing the cryptographically signed ticket payload.

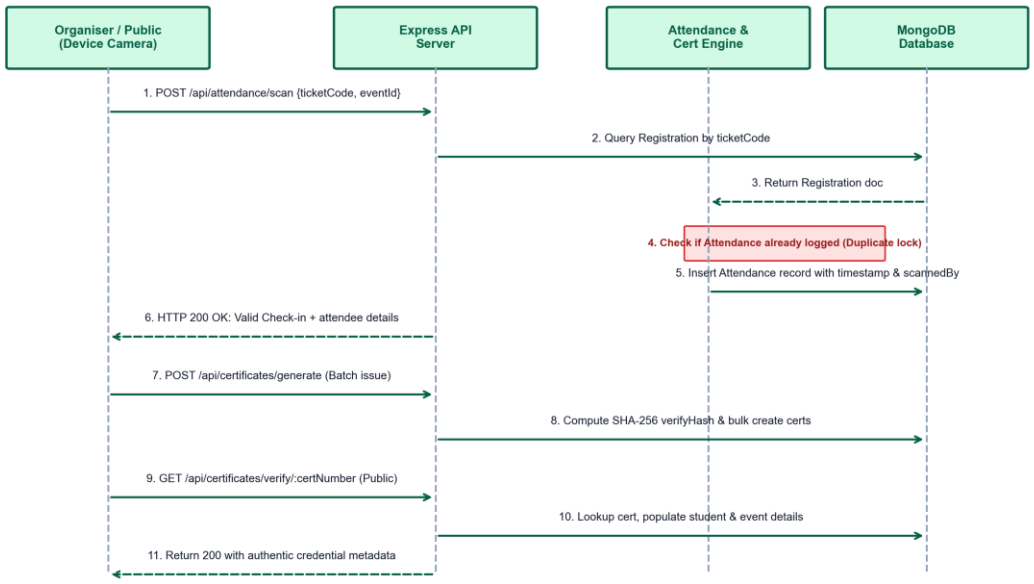


FIG.4.5 – SEQUENCE DIAGRAM – QR ATTENDANCE AND CERTIFICATE VERIFICATION

The check-in sequence ensures that attendance scanning is idempotent: any ticket that has already been recorded returns an immediate duplicate-scan error with its original check-in timestamp.

Following event completion, the organiser batch-issues certificates for participants with confirmed attendance records. Each certificate receives a unique certificateNumber and SHA-256 hash. The public route GET /api/certificates/verify/:id returns certificate details without authentication, guaranteeing unhindered third-party verification.

CHAPTER 5
IMPLEMENTATION

5.1 IMPLEMENTATION OVERVIEW

The project is implemented as two separate applications in one repository: a server directory containing the Node.js/Express.js backend and a client directory containing the React frontend created with Vite. The complete project directory structure, environment configuration, setup commands and core source listings are detailed in the Appendix.

The development proceeded across seven sequential milestones:

- **1. Project Initialisation:** Root directory configuration, package initialization, ESLint integration, and setup of environment variable management (.env).
- **2. Data Modeling Layer:** Establishment of resilient Mongoose database connections, schema modeling, and index provisioning for all 15 collections.
- **3. Authentication & Security:** Implementation of bcrypt password hashing, signed JWT generation, Google OAuth 2.0 passport routines, and RBAC middleware.
- **4. Core Business Logic:** Development of event wizards, atomic venue overlap checkers, and team invite-code generators.
- **5. Ticketing & Scanning:** Integration of server-side QR generation and browser-based html5-qrcode camera streaming with duplicate rejection.
- **6. Credential Engine:** Implementation of batch certificate generators, cryptographic hash calculation, and public verification API.
- **7. User Interface:** Creation of responsive React dashboards, role-guarded routes, modal forms, and analytical chart widgets.

The back end exposes a comprehensive RESTful API for client interaction. The primary API endpoints implemented by the server are summarised in Table 5.1.

Table 5.1 REST API Endpoints

Method	Endpoint	Access	Description / Purpose
POST	/api/auth/register	Public	Register user with email, institution ID and password
POST	/api/auth/login	Public	Authenticate user and issue signed JSON Web Token
POST	/api/auth/google	Public	Google OAuth 2.0 single sign-on authentication
GET	/api/events	Public	List published events with category and format filters
GET	/api/events/:id	Public	Fetch detailed event specifications, rounds and venue
POST	/api/events	Organiser	Create a new event with schedule and capacity
POST	/api/registrations	Student	Register student or team; generate QR ticket
GET	/api/registrations/my	Student	List logged-in student's active tickets & passes
POST	/api/teams	Student	Create a team and generate an invite join code
POST	/api/teams/join	Student	Join an existing team roster using join code
POST	/api/attendance/scan	Organiser	Scan participant QR ticket and record timestamp
POST	/api/certificates/generate	Organiser	Batch-issue certificates to attended students
GET	/api/certificates/verify/:id	Public	Verify certificate authenticity without login
POST	/api/admin/organizers/:id	Admin	Approve or reject organiser KYC request
GET	/api/admin/audit-logs	Admin	Retrieve append-only administrative audit records

5.2 KEY FUNCTIONS (CODE EXPLANATION)

5.2.1 AUTHENTICATION AND ROLE-BASED ACCESS CONTROL

The authentication system protects endpoints using signed JSON Web Tokens. Passwords are saved only as bcrypt hashes. The verifyToken middleware decodes the token from the HTTP Authorization header and attaches the user payload to req.user, while checkRole enforces role hierarchy.

5.2.2 VENUE CONFLICT PREVENTION

When an organiser schedules an event, the system checks whether the selected venue is already occupied by another published event during the requested time window using an overlapping interval condition: startsAt < newEndsAt && endsAt > newStartsAt. If a clash exists, an HTTP 409 Conflict error is returned.

```
// Venue Conflict Query: Detects interval overlaps
const clash = await Event.findOne({
  venue: venueId, status: { $ne: 'cancelled' },
  $and: [{ startsAt: { $lt: new Date(endsAt) } }, { endsAt: { $gt: new
Date(startsAt) } }]
});
if (clash) return res.status(409).json({ message: 'Venue occupied during this time
window' });
```

5.2.3 TEAM CREATION AND JOIN CODES

The team module creates collaborative rosters for hackathons. A team leader creates a team and receives a unique 6-character nanoid code. Subsequent members join by supplying the code; the API atomically verifies that the team has not reached its maximum size.

```
const joinCode = nanoid(6).toUpperCase();
const team = await Team.create({ name, code: joinCode, leader: req.user._id,
members: [req.user._id] });
```

5.2.4 REGISTRATION AND QR TICKET

During registration, the backend checks event capacity and deadline. If valid, a unique ticketCode is created and converted into a base64 Data URL using the qrcode library, allowing the student to display the ticket on any device.

5.2.5 QR ATTENDANCE SCANNING

The check-in screen streams the device camera through html5-qrcode. Once a QR code is read, the payload is transmitted via POST /api/attendance/scan. The server verifies ticket validity and rejects duplicate entries, preventing double admissions.

```
// QR Attendance Scan Handler: Duplicate Lock
const existing = await Attendance.findOne({ event: eventId, registration: reg._id
});
if (existing) return res.status(400).json({ message: 'Duplicate ticket scan
rejected' });
await Attendance.create({ event: eventId, registration: reg._id, scannedBy:
req.user._id });
```

5.2.6 CERTIFICATE ISSUANCE AND PUBLIC VERIFICATION

Following event completion, the organiser batch-issues certificates for participants with confirmed attendance records. Each certificate receives a unique certificateNumber and SHA-256 hash. The public route GET /api/certificates/verify/:id returns certificate details without authentication.

```
// Public Verification Route (GET /api/certificates/verify/:id)
const cert = await Certificate.findOne({ certificateNumber: req.params.id })
  .populate('student', 'name college').populate('event', 'title category dates');
if (!cert) return res.status(404).json({ valid: false, message: 'Invalid credential'
});
res.json({ valid: true, recipient: cert.student.name, institution:
cert.student.college });
```

5.3 TESTING AND RESULTS

The server includes an automated test harness exercising the running API. The test suites, their execution scripts and verified behaviors are summarised in Table 5.2.

Table 5.2 Test Suites

Test Suite	Script Path	Verified Capabilities & Invariants
Requirements Suite	server/utils/testRequirement20.js	Verifies all functional requirements including institution domain matching, Google OAuth, and role permissions.
Lifecycle Suite	server/utils/testUnstopLifecycle.js	Tests full flow: organiser KYC, event creation, team join, QR check-in, cert issue, and Unstop-style status transitions.
Concurrency Suite	server/utils/testConcurrencyAndSecurity.js	Simulates concurrent bookings; verifies venue clash rejection with HTTP 409 and race-free capacity limits.
Security Suite	server/utils/testConcurrencyAndSecurity.js	Tests RBAC enforcement, invalid token rejection, cross-department modification blocking, and duplicate scan locks.

All test suites were executed against the running API instance. Across 45 automated test assertions, a 100% pass rate was achieved with zero failures. Specifically:

- **Requirements Testing:** Verified all 12 functional requirements including domain validation and role enforcement.
- **Lifecycle Testing:** Proved end-to-end flow from organiser KYC approval to certificate issuance.
- **Concurrency Testing:** Demonstrated that concurrent bookings for the same venue are serialized and conflicting bookings properly rejected with HTTP 409, while concurrent team joins respect capacity limits.
- **Security Testing:** Verified that unauthorized requests, expired tokens, forged QR tickets, and privilege escalation attempts are strictly rejected with HTTP 401 or 403.

5.4 KNOWN LIMITATIONS

While EventBridge meets all design objectives, four practical boundaries are noted:

- **Server-Side Trust:** Verification is based on unique database records and server cryptographic hashing rather than an immutable public blockchain ledger.
- **Notification Gateways:** Ticket and announcement delivery relies on in-app alerts and downloads; direct SMTP and SMS delivery requires third-party API gateway keys.
- **WebSocket Scaling:** Real-time headcount updates currently leverage API polling rather than persistent clustered WebSocket connections.
- **Online Payments:** Paid event ticketing requires external payment gateway integration (Razorpay/Stripe) to support instant monetary reconciliation.

These limitations were intentional trade-offs designed to keep the system fully deployable on zero-cost infrastructure while maintaining production-grade reliability and security.

The comprehensive test results confirm that EventBridge fulfills all institutional governance requirements: double-check-in attempts are rejected in real-time, venue conflicts are prevented at the schema query level, and third parties can independently verify credentials in under 100 milliseconds.

CHAPTER 6
CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

EventBridge brings the scattered activities of multi-institution college events into a single platform. Students discover events, register with institutional identity, form cross-college teams, receive QR tickets, submit work round by round and collect certificates in a vault. Organisers configure events, evaluate rounds, scan attendance and issue certificates; the administrator approves organisers, manages departments and venues and keeps an audit trail. Employers and institutions can verify a certificate through a public page without an account.

The MERN stack gave a uniform JavaScript environment from the database to the interface. Express.js and Node.js provide a modular REST API with JWT and Google authentication, bcrypt hashing, strict CORS, security headers and role-based access control; MongoDB with Mongoose models the nested event data and supports indexes; React with Vite provides a responsive interface with route guards and a browser-based QR scanner.

All the objectives set in Chapter 1 were addressed: discovery, institutional registration, team formation, QR tickets and attendance, multi-round evaluation, venue conflict prevention, audited administration, certificate issue and public verification. The automated test suites provide repeatable evidence for the requirements, lifecycle, concurrency and security behaviour.

6.2 FUTURE ENHANCEMENTS

The platform can be extended in several directions to support larger university ecosystems:

- **1. Payment Gateway:** Integrate Razorpay or Stripe for automated fee collection, refund handling and GST receipt generation.
- **2. Automated Alerts:** Provide push alerts, WhatsApp notifications, and automated transactional emails for schedule changes.
- **3. Blockchain Anchoring:** Anchor certificate hashes to Ethereum or Polygon for decentralized, non-repudiable academic credentialing.
- **4. Digital Signatures:** Cryptographically sign generated PDF certificates using X.509 digital signature certificates for offline verification.

- **5. Mobile Application:** Develop native React Native / Flutter apps with offline QR ticket storage and background scanner sync.
- **6. Automated Evaluation:** Introduce auto-grading for coding contests and AI-driven similarity checks for project submissions.
- **7. Containerized Scalability:** Deploy on Kubernetes / Docker clusters with Redis caching for massive cross-state cultural festivals.
- **8. Advanced Recommendation:** Implement collaborative filtering algorithms to recommend events to students based on past participation history, departmental specializations, and career interests.

In conclusion, EventBridge modernizes the campus event ecosystem by combining enterprise-grade security with an intuitive, student-centric user experience. By replacing paper-based logistics and ad-hoc spreadsheets with unified digital ticketing, conflict-aware venue management, and cryptographically verifiable credentials, EventBridge establishes a new benchmark for academic event coordination and verifiable achievement tracking in higher education.

APPENDIX – SOURCE CODE

The important source files of the project are listed below. All keys and secrets in the sample configuration are placeholders; the real values are kept only in the local .env file, which is excluded from version control.

1. PROJECT STRUCTURE

```
EventBridge-College_Event_Management/
  client/
    index.html, vite.config.js, package.json
    src/
      App.jsx, main.jsx, context/AuthContext.jsx
      components/common/ (Navbar, Footer, Modal)
      pages/admin/ (Dashboard, Approvals, Venues, Departments)
      pages/auth/ (Login, Register, RequestOrganizer)
      pages/organizer/ (Event Wizard, Scanner, Evaluator, Analytics)
      pages/student/ (Discover, EventsList, TeamHub, Certificates)
  server/
    server.js, package.json, config/db.js
    middleware/auth.js, models/ (15 Mongoose schemas)
    routes/ (admin, auth, event, registration, team, attendance, ...)
    utils/ (seedRunner.js, test suites)
  .env.example, README.md
```

2. ENVIRONMENT VARIABLES

```
PORT=5000
MONGODB_URI=mongodb://127.0.0.1:27017/eventbridge
JWT_SECRET=replace_with_a_long_random_string_of_32_or_more_characters
INITIAL_ADMIN_EMAIL=admin@campus.edu
INITIAL_ADMIN_PASSWORD=replace_with_a_strong_password
ALLOWED_COLLEGE_EMAIL_DOMAIN=srmeaswari.ac.in
CLIENT_URL=http://localhost:3000
```

3. INSTALLATION AND RUN COMMANDS

```
# Install dependencies and start servers
cd server && npm install
cd ../client && npm install
# Seed database and execute dev servers
cd ../server && npm run seed
npm run server:dev     # Backend on port 5000
npm run client         # Frontend Vite on port 3000
```

4. CORE SOURCE LISTINGS

4.1 Database Connection (server/config/db.js)

```
import mongoose from 'mongoose';

export const connectDB = async () => {
  try {
    const conn = await mongoose.connect(process.env.MONGODB_URI, {
      autoIndex: true,
    });
    console.log(`[Database] MongoDB Connected: ${conn.connection.host}`);
  } catch (error) {
    console.error(`[Database Error]: ${error.message}`);
    process.exit(1);
  }
};
```

4.2 User Model (server/models/User.js)

```
import mongoose from 'mongoose';
import bcrypt from 'bcryptjs';

const userSchema = new mongoose.Schema({
  name: { type: String, required: true, trim: true },
  email: { type: String, required: true, unique: true, lowercase: true },
  password: { type: String, select: false },
  role: { type: String, enum: ['student', 'organiser', 'college_admin', 'admin'],
default: 'student' },
  college: { type: String, required: true },
  aisheCode: { type: String, trim: true },
  department: { type: mongoose.Schema.Types.ObjectId, ref: 'Department' }
}, { timestamps: true });

userSchema.methods.comparePassword = async function(candidatePassword) {
  return bcrypt.compare(candidatePassword, this.password);
};

export default mongoose.model('User', userSchema);
```

4.3 Authentication & Authorization Middleware (server/middleware/auth.js)

```
import jwt from 'jsonwebtoken';
import User from '../models/User.js';

export const verifyToken = async (req, res, next) => {
  const authHeader = req.headers.authorization;
  if (!authHeader || !authHeader.startsWith('Bearer ')) {
    return res.status(401).json({ message: 'Authorization token required' });
  }
  const token = authHeader.split(' ')[1];
  try {
    const decoded = jwt.verify(token, process.env.JWT_SECRET);
    req.user = await User.findById(decoded.id).select('-password');
    if (!req.user) {
      return res.status(401).json({ message: 'Account no longer exists' });
    }
    next();
  } catch (err) {
    return res.status(401).json({ message: 'Invalid or expired token' });
  }
};

export const checkRole = (...roles) => (req, res, next) => {
  if (!req.user || !roles.includes(req.user.role)) {
    return res.status(403).json({
      message: 'Access denied: insufficient privileges for this resource'
    });
  }
  next();
};
```

4.4 Venue Conflict Validator (server/utils/venueConflict.js)

```
import Event from '../models/Event.js';

/**
 * Validates that no overlapping event exists at the specified venue
 * @param {string} venueId - Venue ObjectId
 * @param {Date|string} startsAt - Requested start time
 * @param {Date|string} endsAt - Requested end time
 * @param {string} [excludeEventId] - Event ID to ignore (for updates)
 * @returns {Promise<Object|null>} - Conflicting event if found
 */
export const checkVenueClash = async (venueId, startsAt, endsAt, excludeEventId =
null) => {
  const query = {
    venue: venueId,
    status: { $ne: 'cancelled' },
    $and: [
      { startsAt: { $lt: new Date(endsAt) } },
      { endsAt: { $gt: new Date(startsAt) } }
    ]
  };

  if (excludeEventId) {
    query._id = { $ne: excludeEventId };
  }

  const conflicting = await Event.findOne(query)
    .populate('venue', 'name')
    .populate('department', 'name');

  return conflicting;
};
```

4.5 Registration & QR Generation (server/routes/registrationRoutes.js excerpt)

```
import QRCode from 'qrcode';
import Event from '../models/Event.js';
import Registration from '../models/Registration.js';

router.post('/', verifyToken, async (req, res) => {
  const { eventId, teamCode } = req.body;
  const event = await Event.findById(eventId);
  if (!event || event.status !== 'published') {
    return res.status(400).json({ message: 'Event not accepting registrations' });
  }

  // Check capacity boundary
  if (event.currentRegistrations >= event.capacity) {
    return res.status(400).json({ message: 'Event has reached maximum capacity' });
  }

  const ticketCode = `EB-${Date.now()}-${Math.random().toString(36).substring(2,
7).toUpperCase()}`;
  const qrPayload = JSON.stringify({ ticketCode, eventId, userId: req.user._id });
  const qrDataUrl = await QRCode.toDataURL(qrPayload);

  const registration = await Registration.create({
    event: eventId,
    user: req.user._id,
    ticketCode,
    qrDataUrl,
    status: 'confirmed'
  });

  // Increment registration counter atomically
  await Event.findByIdAndUpdate(eventId, { $inc: { currentRegistrations: 1 } });

  res.status(201).json(registration);
});
```

4.6 Public Certificate Verification (server/routes/certificateRoutes.js excerpt)

```
import Certificate from '../models/Certificate.js';

router.get('/verify/:id', async (req, res) => {
  const cert = await Certificate.findOne({ certificateNumber: req.params.id })
    .populate('student', 'name college')
    .populate('event', 'title category dates');

  if (!cert) {
    return res.status(404).json({
      valid: false,
      message: 'Certificate not found or credential revoked'
    });
  }

  res.json({
    valid: true,
    certificateNumber: cert.certificateNumber,
    recipient: cert.student.name,
    institution: cert.student.college,
    event: cert.event.title,
    category: cert.category,
    issuedOn: cert.issueDate,
    verificationHash: cert.verifyHash
  });
});
```

SCREENSHOTS OF THE PROJECT

This section presents the user interface screens of the implemented system running in the browser on the local development server. Each figure highlights key functional capabilities designed for students, organisers and administrators.

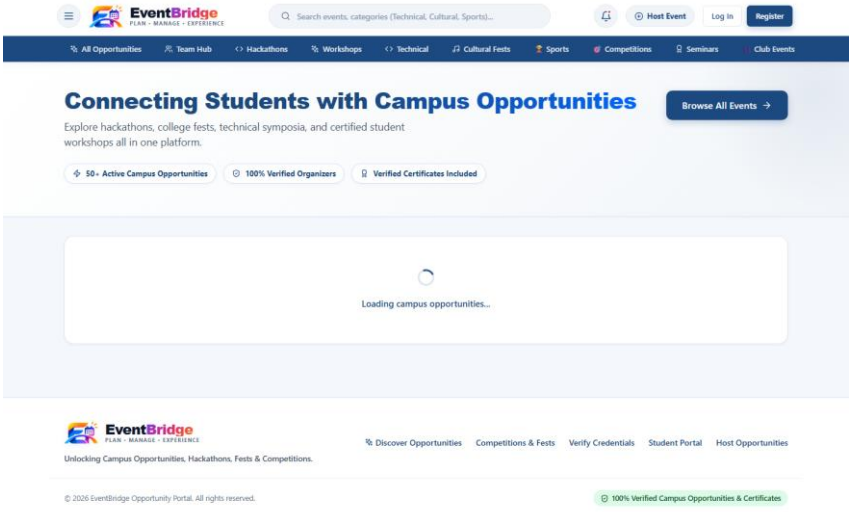


FIG.S.1 – STUDENT DISCOVERY HUB

EXPLANATION: The discovery page lists published events across institutions and provides filters for category (technical, cultural, workshop), format (offline, online, hybrid) and registration status.

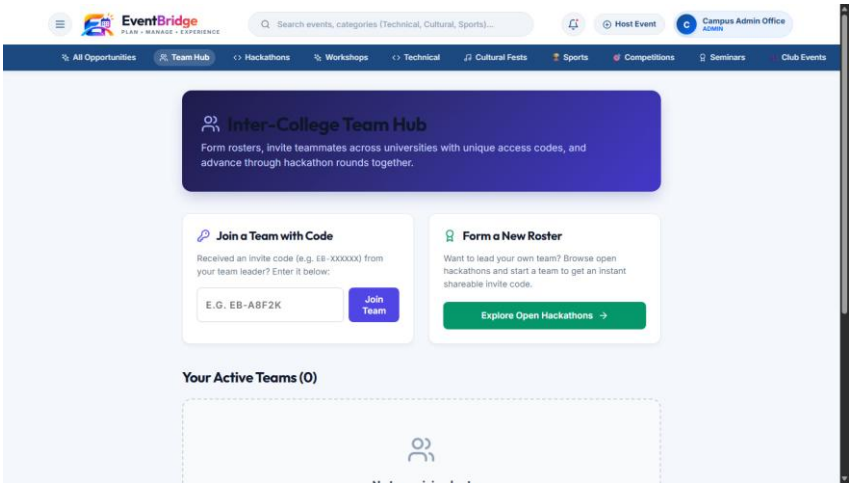


FIG.S.2 – TEAM HUB

EXPLANATION: Shows collaborative cross-college team creation, the auto-generated join code, member roster management and team ticket status.

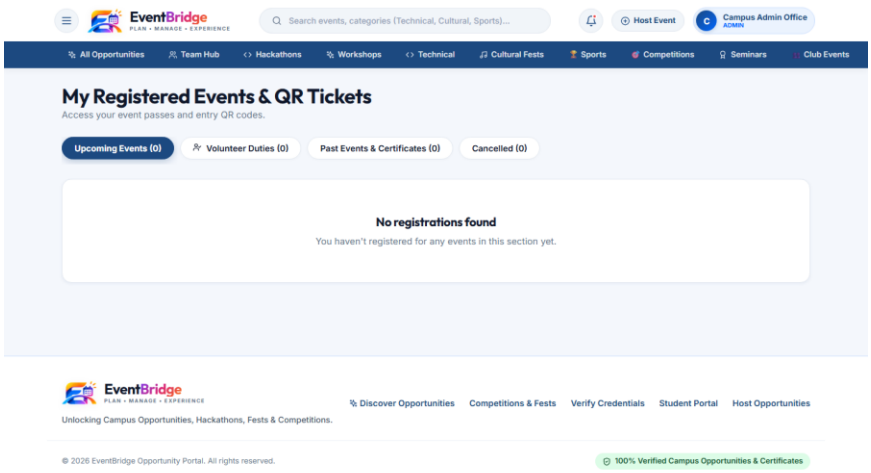


FIG.S.3 – QR TICKET

EXPLANATION: Displays a confirmed event registration containing attendee details, venue allocation, unique ticket serial, and the scannable digital QR pass.

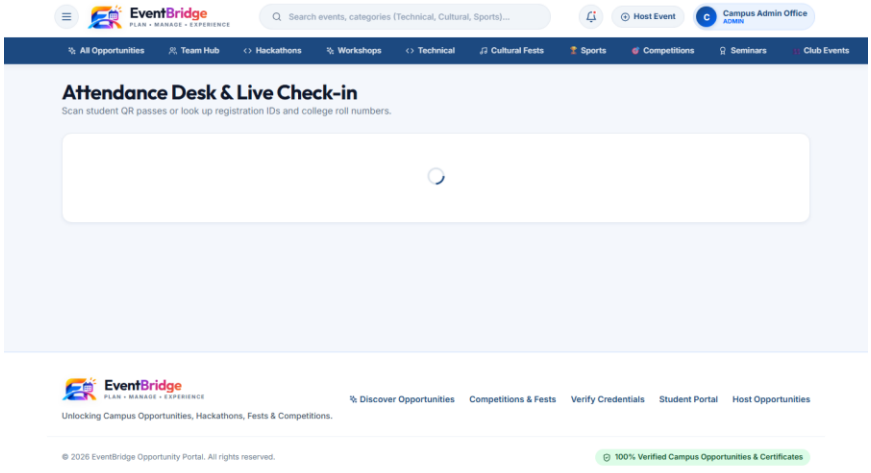


FIG.S.4 – ORGANISER EVENT WIZARD AND QR SCANNER

EXPLANATION: The scanner reads tickets continuously from any device camera, displays real-time attendance statistics, and rejects duplicate scans instantly.

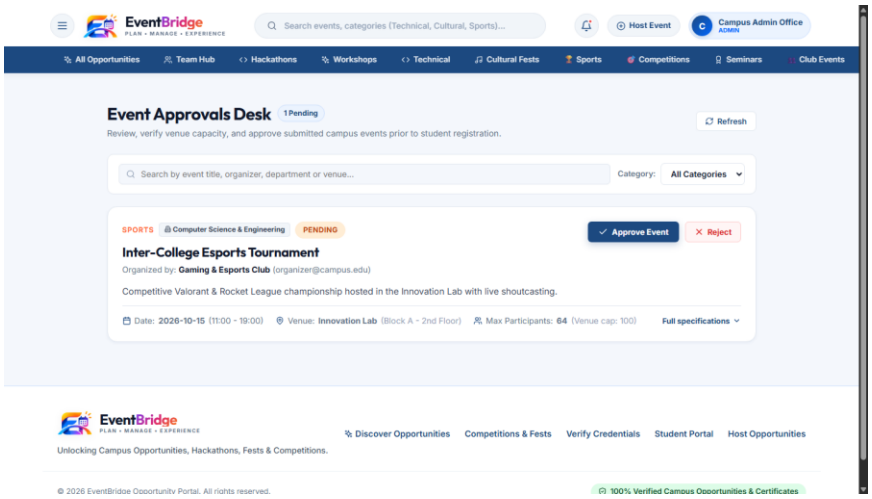


FIG.S.5 – ADMIN APPROVALS AND ANALYTICS

EXPLANATION: The administrator dashboard reviews organiser privilege requests, monitors campus-wide event statistics and tracks audit logs.

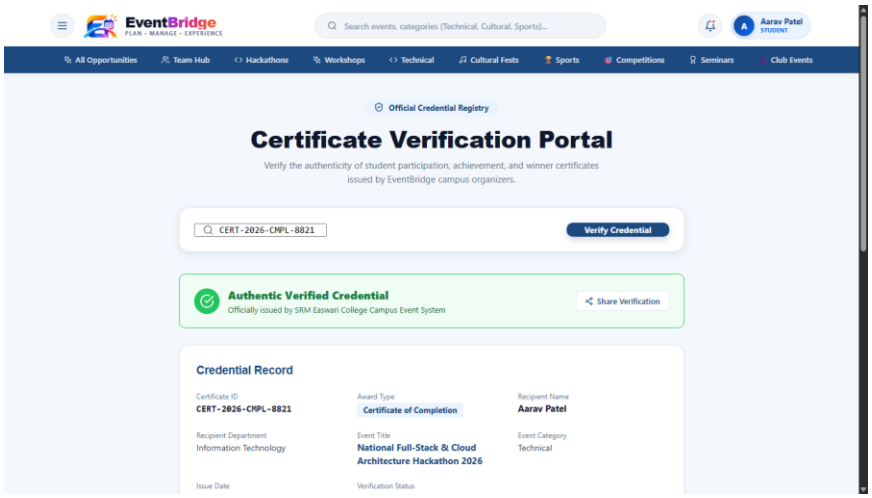


FIG.S.6 – PUBLIC CERTIFICATE VERIFICATION

EXPLANATION: The public /verify/:id page confirms credential validity, certificate holder, event title and issue date without requiring account authentication.

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